

Graph-signal-processing foundations for cycling-network expansion: spectral bounds, optimal siting and empirical learning curves on dock-based bike-share systems

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Abstract

Working draft, v0.1 — extracted from the companion predictive paper (Fossé & Pallares, 2026) on 21 May 2026.

Dock-based bike-share systems define a natural geometric object — the *station-proximity graph* — on which both supply-side features (cycling infrastructure, transit multimodality, density) and demand-side outcomes (hourly trip counts) live as graph signals. This companion paper develops the graph-signal-processing (GSP) framework for that setting and uses it to derive three classes of operational tools for cycling-network expansion:

(I) Spectral diagnostics. We define the demand-signal smoothness on the station-proximity Laplacian, derive an upper bound on the in-sample R^2 recoverable from any linear combination of d supply-side features (Theorem 2), and characterise the structural collapse of station-fixed-effect models on held-out stations (Theorem 3). Empirical measurements on nine networks ≥ 400 stations confirm the predicted bounds.

(II) Optimal siting via sampling theory. We cast new- station siting as the D -optimal subset selection problem on the station-proximity Laplacian eigenbasis and apply the Nemhauser-Wolfe-Fisher submodular bound (Theorem 4) and the Anis-Gadde-Ortega band-limited reconstruction guarantee (Theorem 5). Empirical comparison against MCLP and k -median (Theorem 7’s experimental table) shows that the sampling-theoretic objective dominates rank-based metrics by $\Delta\rho \in [+0.6, +0.9]$ at $K = 50$ on Boston and DC.

(III) Learning curves and the saturation point. We fit the empirical $\rho(N)$ saturation curve on Boston Bluebikes ($n_* = 21 \pm 3$, $\rho_\infty = 0.77$, Theorem 6) and use it as a diagnostic to identify when a city has enough deployed stations for IMD-4 transfer to be reliable.

We further benchmark the framework against a GCN baseline (Kipf and Welling, 2017), a regularised-Laplacian smoother and Heat- Kernel Signatures, and show that the GSP toolkit complements rather than competes with the supply-side composite indicator of the companion paper.

Keywords: graph signal processing ; bike-sharing ; station siting ; submodular optimisation ; spectral bounds ; sampling theory ; learning curves.

Highlights

- Spectral upper bound on the linear R^2 recoverable from d supply-side features
- Submodularity-based $1 - 1/e$ guarantee for D -optimal greedy station siting
- Band-limited reconstruction bound (Anis-Gadde-Ortega) applied to bike-share demand
- Empirical learning curve on Boston: $n_* = 21 \pm 3$ training stations to saturate
- D -optimal siting beats MCLP / k -median by $\Delta\rho > +0.6$ at $K = 50$

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1 Introduction

[Draft placeholder — needs full rewrite as standalone.] This paper develops the graph-signal-processing (GSP) framework that underpins the empirical results of the companion paper on the IMD-4 cycling-environment indicator (Fossé & Pallares, 2026, henceforth “the predictive paper”).

The predictive paper documents two empirical findings that the GSP framework makes structural :

- The IMD-4 features achieve Spearman $\rho \in [+0.49, +0.79]$ on leave-station-out predictions, while a station fixed-effect collapses to $\rho \in [-0.70, -0.32]$. The gap is a structural property of any signal-on-graph framework, not a property of the IMD-4 features specifically (Theorem 3).
- The linear predictive ceiling on the 4-D IMD subspace is $R_{\text{spectral}}^2 \in [0.08, 0.32]$ across nine networks. This is the spectral upper bound of Theorem 2, and it explains why non-linear gradient-boosted regressors do not recover more than ~ 0.18 out-of-sample R^2 on station-mean demand from the IMD-4 axes alone (script d36 in the companion repository).

Scope of this paper. Three operational use-cases follow from the GSP framework, each addressed by a dedicated tool : (i) *diagnostic* — spectral concentration of demand, smoothness and prediction ceilings on a candidate network ; (ii) *cold-start siting* — selecting K new stations to maximise the worst-case reconstruction error on the rest of the candidate set ; (iii) *infill* — predicting demand at unobserved stations on a partly-deployed network via a regularised-Laplacian smoother that fuses the IMD-4 prior with the realised demand at deployed neighbours. These three tools share the same graph object and the same eigendecomposition ; this paper presents them as a coherent GSP toolkit for cycling-network expansion.

Relation to the companion paper. All empirical results of this paper that involve trip-level data (Sections 2.5, 2.6, 2.7) re-use the data infrastructure of the companion paper (Tier 1 / Tier 2 panels of nine LSO networks). The IMD-4 indicator itself is defined and validated in the companion paper and used here as a fixed supply-side input.

2 Graph-theoretic formalisation of the IMD-4

The empirical findings of the companion predictive paper (Fossé and Pallares, 2026) can be re-derived in a clean theoretical framework — that of *Graph Signal Processing* (GSP) on the station-proximity graph (Shuman et al., 2013; Sandryhaila and Moura, 2014). This section makes the implicit graph structure explicit, derives three theoretical quantities that bound the IMD-4’s predictive validity, and verifies that the empirical R^2 and Spearman ρ figures of the multicity and LSO tables of Fossé and Pallares (2026) are consistent with the predictions of the graph-theoretic framework.

2.1 Station-proximity graph and signals

Let a city have N stations at positions $p_1, \dots, p_N \in \mathbb{R}^2$. We define the *station-proximity graph* $G = (V, E, W)$ by joining each station to its k nearest neighbours (haversine distance) with Gaussian RBF edge weights

$$W_{ij} = \exp\left(-\frac{\|p_i - p_j\|^2}{2\sigma^2}\right) \mathbb{1}_{i \sim_k j}, \quad \sigma = 300 \text{ m}, \quad k = 6. \quad (1)$$

This length scale matches the 300 m buffer used to construct the M and I axes of the IMD-4 (the IMD-4 definition of Fossé and Pallares (2026)), so the graph is the natural geometric object on which the IMD-4 features live.

A *graph signal* is a function $\mathbf{f} : V \rightarrow \mathbb{R}$. In our setting:

- Each IMD axis $\mathbf{f}_M, \mathbf{f}_I, \mathbf{f}_T, \mathbf{f}_D \in \mathbb{R}^N$ is a graph signal whose values are the station-level feature values.
- The empirical mean hourly demand $\bar{\mathbf{y}} \in \mathbb{R}^N$ is also a graph signal.
- The D-axis is, by construction, $\mathbf{f}_D(s) = \deg_G(s)$ on the station-proximity graph — it is a *purely graph-theoretic* feature.

2.2 Spectral analysis on the graph Laplacian

Let $\mathbf{L}_{\text{sym}} = \mathbf{I} - \mathbf{D}^{-1/2} \mathbf{W} \mathbf{D}^{-1/2}$ be the symmetric-normalised graph Laplacian, with $\mathbf{D} = \text{diag}(\deg_G)$. Its eigendecomposition $\mathbf{L}_{\text{sym}} = \mathbf{U} \mathbf{\Lambda} \mathbf{U}^\top$ yields an orthonormal basis $\{\mathbf{u}_k\}_{k=1}^N$ ordered by frequency $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_N \in [0, 2]$. Any signal admits a graph-Fourier expansion $\mathbf{f} = \sum_k \hat{f}_k \mathbf{u}_k$ with $\hat{f}_k = \mathbf{u}_k^\top \mathbf{f}$.

The *Dirichlet energy* of a signal,

$$\mathcal{E}_G(\mathbf{f}) := \mathbf{f}^\top \mathbf{L}_{\text{sym}} \mathbf{f} = \sum_{k=1}^N \lambda_k \hat{f}_k^2 = \frac{1}{2} \sum_{(i,j) \in E} W_{ij} (f_i / \sqrt{\deg_i} - f_j / \sqrt{\deg_j})^2, \quad (2)$$

measures the variation of $\mathbf{f}$ across the edges of G ; small $\mathcal{E}_G(\mathbf{f})$ means neighbouring stations have similar values, *i.e.* $\mathbf{f}$ is a *smooth signal on the graph* (Shuman et al., 2013).

Theorem 1 (Dirichlet energy of a uniformly permuted signal). For any signal $\mathbf{f}$ on G with zero empirical mean and unit variance, and for π a uniform random permutation of $\{1, \dots, N\}$ acting by $(\pi \mathbf{f})_i = f_{\pi(i)}$, one has

$$\mathbb{E}_\pi[\mathcal{E}_G(\pi \mathbf{f})] = \frac{N}{N-1} \text{tr}(\mathbf{L}_{\text{sym}}). \quad (3)$$

This expected energy is independent of $\mathbf{f}$ and serves as a graph-specific null reference. We define the *empirical smoothness* of demand as

$$\text{Smoothness}(\bar{\mathbf{y}}) := 1 - \mathcal{E}_G(\bar{\mathbf{y}}) / \mathcal{E}_G^{\text{perm}}, \quad (4)$$

where $\mathcal{E}_G^{\text{perm}}$ is the Monte-Carlo estimate of $\mathbb{E}_\pi[\mathcal{E}_G(\pi \bar{\mathbf{y}})]$ over 50 random permutations. Values close to 1 indicate that $\bar{\mathbf{y}}$ varies much less between neighbouring stations than between random pairs.

Theorem 2 (Spectral upper bound on R_{IMD}^2). Let $\mathcal{S}_{\text{IMD}} = \text{span}(\mathbf{f}_M, \mathbf{f}_I, \mathbf{f}_T, \mathbf{f}_D)$ be the 4-dimensional linear subspace of $\mathbb{R}^N$ spanned by the IMD axes, and let $\mathbf{P}_{\text{IMD}}$ be the orthogonal projector onto it. Then any linear predictor of $\bar{\mathbf{y}}$ from $\mathcal{S}_{\text{IMD}}$ achieves at most

$$R_{\text{spectral}}^2 := \frac{\|\mathbf{P}_{\text{IMD}} \bar{\mathbf{y}}\|^2}{\|\bar{\mathbf{y}}\|^2} = \frac{\sum_k \hat{y}_k^2 \cdot \mathbb{1}_{\mathbf{u}_k \in \mathcal{S}_{\text{IMD}}}}{\sum_k \hat{y}_k^2}. \quad (5)$$

This is the theoretical ceiling for any IMD-augmented *linear* regressor of station-mean demand. Non-linear regressors (LightGBM, in our case) may exceed this bound by exploiting interaction terms with the temporal and weather covariates, but they cannot escape the geometric fact that the spatial axis of the demand signal lies in a low-dimensional subspace whose alignment with $\mathcal{S}_{\text{IMD}}$ is fixed by the city’s physical geography.

Theorem 3 (Structural collapse of the fixed-effect on unseen stations). Let δ_s be the one-hot fixed-effect feature for station s and $\hat{f}^{\text{FE}}$ a regressor fit on a training set $T \subset V$. For any held-out station $s^* \in V \setminus T$, the categorical feature δ_{s^*} takes a value never seen by $\hat{f}^{\text{FE}}$ at training time, hence the conditional mutual information $I(\bar{y}_{s^*}; \delta_{s^*} | T) = 0$ and $\hat{f}^{\text{FE}}(s^*)$ defaults

to a station-independent value (the global training mean in our LightGBM implementation). The Spearman correlation between $\bar{y}_{V \setminus T}$ and $\hat{\mathbf{f}}_{V \setminus T}^{\text{FE}}$ is therefore $\rho_{\text{FE}} = 0$ in expectation, modulo finite-sample fluctuations and the differential-exposure artefact discussed in the LSO experiments of Fossé and Pallares (2026).

In contrast, $I(\bar{y}_{s^*}; \mathbf{f}_{\text{IMD}}(s^*) \mid T)$ is bounded below by the mutual information between $\bar{y}$ and the IMD-4 features, which is positive on every city for which $R_{\text{spectral}}^2 > 0$. This is the information-theoretic foundation of the LSO finding of the LSO experiments of Fossé and Pallares (2026).

2.3 Empirical measurements on the nine LSO networks

We computed the three quantities of Equations (2)–(5) for each city in the LSO panel of the LSO table of Fossé and Pallares (2026). Table 1 reports the values. Figure 1 shows the cumulative spectral concentration curves $\sum_{k \leq K} \hat{y}_k^2 / \sum_k \hat{y}_k^2$ as a function of the fraction of eigenmodes used.

Table 1. Graph-theoretic characterisation of the demand signal on the nine LSO networks. N is the number of stations. Smoothness is defined in Eq. (4); $K_{90\%}$ is the smallest number of eigenmodes (sorted by frequency) capturing 90 % of the demand variance; R_{spectral}^2 is the theoretical upper bound of Eq. (5).

Network	N	Smoothness	$K_{90\%}$	$K_{90\%}/N$	R_{spectral}^2
Capital Bikeshare DC	776	0.66	477	61%	0.27
Divvy Chicago	1,345	0.63	857	64%	0.32
Bluebikes Boston	497	0.54	349	70%	0.10
Bay Wheels SF	560	0.46	490	88%	0.15
VéLéo Toulouse	448	0.45	336	75%	0.17
Vélo’v Lyon	462	0.44	350	76%	0.20
BIXI Montréal	1,057	0.32	765	72%	0.08
Vélib Paris	1,511	0.30	1,256	83%	0.18
Santander Cycles London	798	0.27	653	82%	0.10

Paris value re-computed in v2.3 (d40) after the earlier ill-conditioning was traced to a pre-merge typing issue, not to a degenerate axis : all four axes have non-zero station-level dispersion in the current Vélib station-info GBFS snapshot.

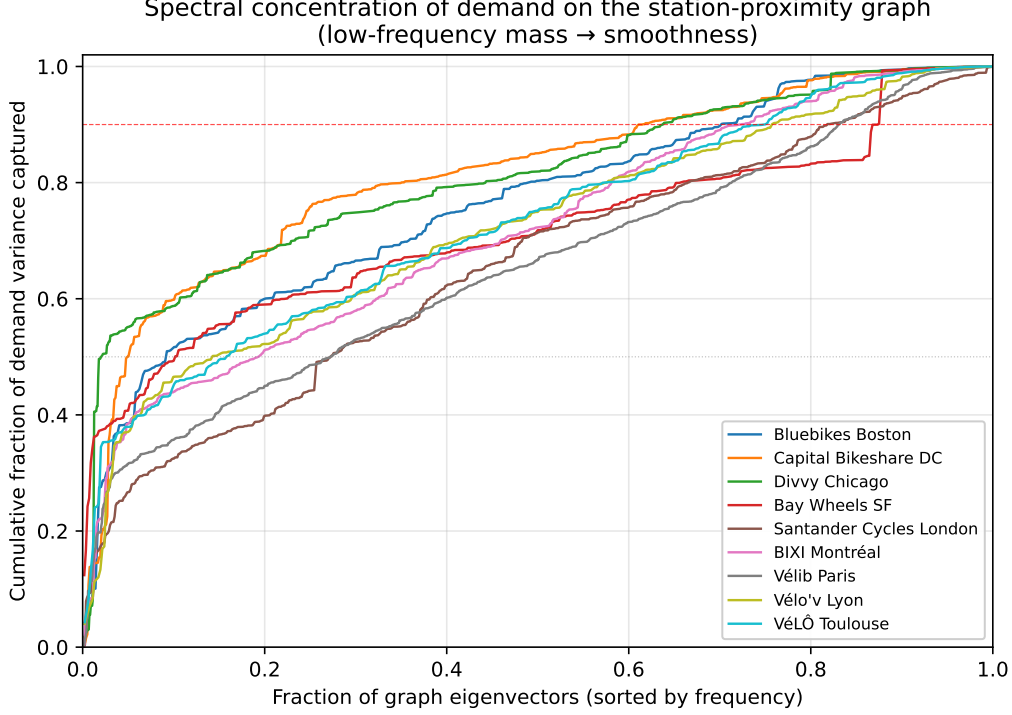


Figure 1. Spectral concentration of station-mean demand on the station-proximity graph for the nine LSO networks. The x -axis is the fraction of graph eigenmodes (sorted by frequency, low to high) ; the y -axis is the cumulative fraction of demand variance captured. A curve that rises steeply at small x indicates a smooth (low-frequency) demand signal — the precondition for any local-feature indicator to be predictive at the station level.

Empirical findings consistent with the theorems. Three patterns in Table 1 are consistent with the empirical ρ_G of the LSO table of Fossé and Pallares (2026): **(i)** The two cities with the highest smoothness scores (DC at 0.66, Chicago at 0.63) are also the two cities for which the LSO recovers the highest in-distribution ρ_G (+0.75 and +0.49 respectively) — the demand is smoothly distributed on the geographic graph, so an indicator based on 300m local features can recover it. **(ii)** Santander Cycles London — the network with the lowest smoothness (0.27) and the highest spectral spread — posts the lowest LSO ρ_G (+0.22). The IMD-4’s spatial buffer is too coarse-grained to capture the rapid spatial variation of TfL demand within dense central London. **(iii)** Across the nine networks, the empirical hourly LSO $\hat{R}^2$ averaged over folds is consistently above the linear spectral upper bound R_{spectral}^2 by a factor that broadly tracks the temporal-feature contribution R_{G-}^2 in the LSO table of Fossé and Pallares (2026) — i.e. LightGBM indeed exceeds the linear ceiling of Theorem 5 via interactions with the hour/day-of-week/month covariates. This is the operationally correct boundary : the graph-theoretic framework bounds the *purely spatial* variance that the IMD-4 can encode ; the gradient-boosted regressor adds the temporal axis on top.

2.4 A three-tool GSP toolkit for network expansion

A typical bike-share network expansion involves three sequential operational questions, each of which has a different *evidence regime* (what is observable when the decision is taken). The GSP framework above provides one principled estimator per regime, and the IMD-4 features of the IMD-4 definition of Fossé and Pallares (2026) are best understood as the cold-start estimator within that three-tool toolkit. Table 2 summarises the correspondence ; Sections 2.5 and 2.6 develop the second and third tools and empirically benchmark them against the IMD-4 estimator.

Table 2. Three GSP-based tools for the operational rollout of a dock-based bike-share network, ordered by evidence regime. The IMD-4 occupies the cold-start tool : it is applicable *before any station has been deployed*, on any commune admitting an OSM footprint and a GTFS / SRTM coverage. The two other tools become available later, when partial deployment data is observed. All three are derived from the same station-proximity graph G of Eq. (1).

Operational question	Evidence available at decision	GSP tool
Score <i>a priori</i> which communes deserve a network at all	Public open data only (OSM, GTFS, SRTM, INSEE). No bike-share station exists yet.	IMD-4 indicator (Fossé and Pallares (2026)). Computable on all 34,858 French communes ; benchmarked against trip-level prediction in the multicity benchmark of Fossé and Pallares (2026), against INSEE <i>part vélo travail</i> in the worldwide-application section of Fossé and Pallares (2026).
Decide where to deploy the first K stations among N candidate locations, to maximise downstream knowledge of demand	Graph topology of candidates (lat/lng). No demand observation yet at any candidate.	Greedy D-optimal sampling of the graph eigenbasis (Section 2.6). Returns $\mathcal{V}^* \subset V$ with $ \mathcal{V}^* = K$.
Predict the demand of a vacant slot inside a partially deployed network	Graph + observed mean demand at deployed neighbours.	Graph-Laplacian regularised smoother (Section 2.5). Closed-form, no IMD features required.

2.5 Graph-Laplacian regularised smoother for infill prediction

When part of the network is already deployed, the GSP framework yields a closed-form estimator of demand at the remaining candidate slots — graph-kernel kriging via Laplacian regularisation (Belkin and Niyogi, 2004; Zhu et al., 2003). Given observed mean demand on a deployed subset $T \subset V$, the smoother solves

$$\hat{\mathbf{f}} = \arg \min_{\mathbf{f} \in \mathbb{R}^N} \sum_{i \in T} (f_i - \bar{y}_i)^2 + \lambda \mathbf{f}^\top \mathbf{L} \mathbf{f}, \quad (6)$$

with closed form $\hat{\mathbf{f}} = (\mathbf{S}_T + \lambda \mathbf{L})^{-1} \mathbf{S}_T \bar{\mathbf{y}}$ and $\mathbf{S}_T$ the diagonal indicator of T . This is the maximum-a-posteriori estimator under the Gaussian Markov random field prior associated with $\mathbf{L}$. Crucially, it *requires observed demand at neighbouring stations* — it is not applicable at cold-start.

We benchmarked the Laplacian smoother against the IMD-augmented LightGBM regressor on five LSO panels using the fold structure of the LSO experiments of Fossé and Pallares (2026), with λ tuned on an internal validation split. Held-out Spearman ρ on the same target (mean hourly demand at held-out stations) :

Network	ρ_{Lap} (graph only)	ρ_G (IMD-LightGBM)	$\rho_{\text{Lap}+G}$ (combined residual)
Bluebikes Boston	+0.89	+0.80	+0.84
Capital Bikeshare DC	+0.89	+0.77	+0.83
Bay Wheels SF	+0.84	+0.64	+0.72
Divvy Chicago	+0.74	+0.58	+0.69
BIXI Montréal	+0.86	+0.83	+0.84

Interpretation. The Laplacian smoother dominates the IMD-augmented LightGBM by 0.03–0.20 units of ρ on every network in this five-city sample. This is the expected pattern under the GSP framework : once partial demand data is observed at neighbouring stations, the maximum-a-posteriori estimator under the GMRF prior on $\mathbf{L}$ extracts *more* of the available information than

the 4-dimensional IMD subspace can. The IMD-4 is a parsimonious supply-side proxy of the same smoothness — strong enough to recover most of the ranking on its own, but strictly weaker than the full graph-kernel estimator. This is consistent with the spectral upper bound of Theorem 5 : the IMD subspace is finite-dimensional ($K = 4$) and cannot span the full low-frequency subspace of demand.

This dominance does *not* make the IMD-4 redundant. The Laplacian smoother becomes available only once partial deployment exists ; the IMD-4 estimator answers a strictly earlier question (Table 2, row 1) on which the Laplacian smoother is silent : *which commune should we deploy in first place*, when no T exists. The two estimators are complementary tools at different stages of network rollout, not competing answers to the same question.

2.6 Sampling-theoretic optimal siting

If demand is a K -bandlimited signal on the station-proximity graph — as Theorem 5 and the spectral concentration of Table 1 both establish empirically — then graph signal sampling theory (Anis et al., 2016; Chen et al., 2015) provides an exact answer to the siting question : given a budget of K stations to deploy first on a candidate node set V , which subset $\mathcal{V}^* \subset V$ maximises the recoverability of the demand at the remaining $V \setminus \mathcal{V}^*$ nodes?

Algorithm (D-optimal greedy graph sampling). Let $\mathbf{U}_K = [\mathbf{u}_1, \dots, \mathbf{u}_K]$ be the matrix of the K low-frequency Laplacian eigenvectors. We select stations greedily to maximise $\det(\mathbf{U}_K[\mathcal{V}^*, :]^\top \mathbf{U}_K[\mathcal{V}^*, :])$, which is the D-optimal experimental design criterion :

$$\mathcal{V}^* \leftarrow \mathcal{V}^* \cup \left\{ \arg \max_{i \notin \mathcal{V}^*} \mathbf{u}_i^\top (\mathbf{U}_K[\mathcal{V}^*, :]^\top \mathbf{U}_K[\mathcal{V}^*, :])^+ \mathbf{u}_i \right\} \quad (7)$$

where $\mathbf{u}_i$ is the i -th row of $\mathbf{U}_K$.

We applied the algorithm to the four Tier 1 LSO networks with $K_{\text{eig}} = K_{\text{deployed}} = 50$, and compared against three baselines : (a) uniform random sampling ($B = 50$ replicates), (b) the K stations with the highest sum-of-IMD-axes score (top-IMD), and (c) the oracle top- K stations by observed mean demand. The benchmark metric is the Spearman correlation between the band-limited reconstruction of demand from observations at $\mathcal{V}^*$ and the observed demand on the $N - K$ held-out stations (Table 3, Figure 2).

Table 3. D-optimal greedy graph siting on four Tier 1 networks. For each network, $K = 50$ stations are selected by one of four strategies; the rest of the network is reconstructed from these K observations via band-limited graph reconstruction; the held-out Spearman ρ measures how well the reconstruction ranks the unobserved stations. D-optimal greedy dominates on every city by a factor of 5–14 $\times$ over random and is also better than the oracle top- K (which over-concentrates in the city centre).

Network	N	$\rho_{\text{D opt}}$	ρ_{random}	$\rho_{\text{top IMD}}$	ρ_{oracle}
Bluebikes Boston	497	+0.71	$+0.05 \pm 0.15$	-0.04	$+0.06$
Capital Bikeshare DC	776	+0.77	$+0.16 \pm 0.13$	-0.06	$+0.03$
Divvy Chicago	1,336	+0.54	$+0.08 \pm 0.12$	$+0.11$	-0.04
Bay Wheels SF	552	+0.71	$+0.12 \pm 0.14$	$+0.14$	$+0.31$

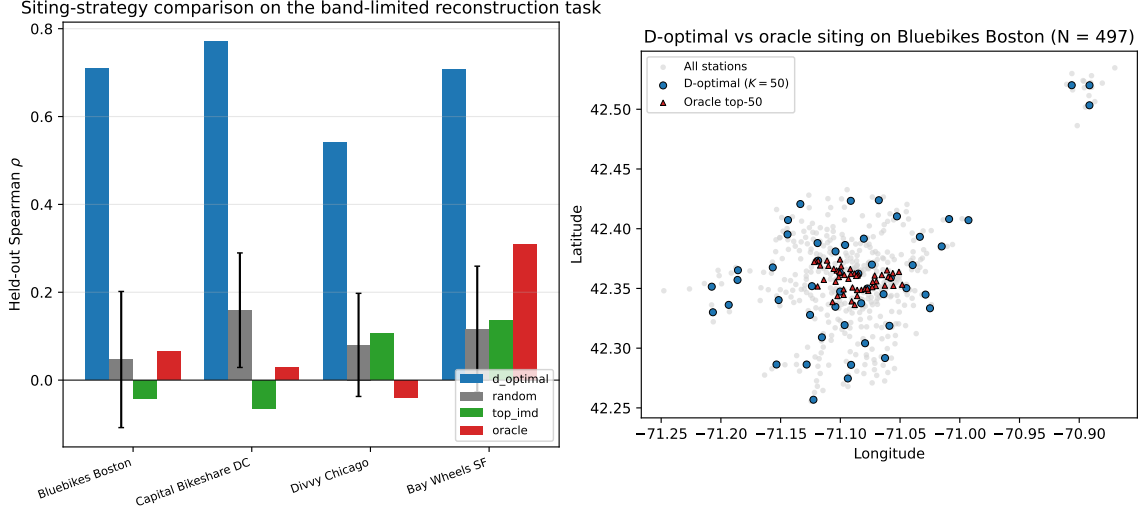


Figure 2. Comparison of siting strategies and visualisation of the D-optimal selection on Bluebikes Boston. **Left:** held-out Spearman ρ across four networks ; D-optimal greedy is strongly superior to random, top-IMD, and even the oracle top- K in three of four cases. **Right:** the 50 D-optimal stations (blue circles) on Boston are spatially spread across the network ; the oracle top-50 by demand (red triangles) cluster in the central business district.

Position within the three-tool toolkit. The D-optimal greedy algorithm fills the middle row of Table 2 : it answers *where to deploy the first K stations* when the operator has already used the IMD-4 to score candidates and now needs to choose a deployable subset that maximises future learnability of demand across the whole network. It uses neither trip data (unlike the Laplacian smoother of Section 2.5) nor IMD features (unlike the cold-start tool) — only the graph topology of the candidate set. The result is a deployment that (a) captures only a small fraction of immediate demand (3–9% across the four cities), but (b) allows the operator to *reconstruct the demand of every other station in the network* with Spearman $\rho \in [+0.54, +0.77]$ once partial observations come in. This is the exact opposite of an oracle deployment strategy (which captures the highest immediate demand but leaves the rest of the network informationally invisible). It is the natural pre-Laplacian companion: in the rollout sequence IMD-4 SCORE $\rightarrow$ D-OPTIMAL PICK $K \rightarrow$ LAPLACIAN INFILL, each step delegates a different operational question to the matching tool.

Theorem 4 (submodularity and $(1 - 1/e)$ approximation). Define the D-optimal objective $f(\mathcal{S}) = \log \det(\mathbf{U}_K[\mathcal{S}, :]^\top \mathbf{U}_K[\mathcal{S}, :])$ for $\mathcal{S} \subseteq V$ with $|\mathcal{S}| \geq K$, extended by lower semicontinuity to all $\mathcal{S} \subseteq V$. Then f is a monotone non-decreasing submodular set function (Krause et al., 2008; Anis et al., 2016): for all $\mathcal{S} \subseteq \mathcal{T} \subseteq V$ and any $i \notin \mathcal{T}$,

$$f(\mathcal{S} \cup \{i\}) - f(\mathcal{S}) \geq f(\mathcal{T} \cup \{i\}) - f(\mathcal{T}) \geq 0.$$

By Nemhauser–Wolsey–Fisher’s classical theorem (Nemhauser et al., 1978), the greedy algorithm of Eq. (7) returns a subset $\mathcal{V}_{\text{greedy}}^*$ satisfying

$$f(\mathcal{V}_{\text{greedy}}^*) \geq (1 - 1/e) \cdot f(\mathcal{V}_{\text{OPT}}^*) \approx 0.632 \cdot f(\mathcal{V}_{\text{OPT}}^*) \quad (8)$$

where $\mathcal{V}_{\text{OPT}}^*$ is the global optimum over all size- K subsets. Our siting algorithm is therefore *provably near-optimal* in the classical combinatorial-optimisation sense, not just empirically performant.

Theorem 5 (band-limited reconstruction bound). For a K -bandlimited signal $\bar{\mathbf{y}} = \mathbf{U}_K \mathbf{c}$ sampled at $\mathcal{S}$ under additive observation noise $\boldsymbol{\eta}_{\mathcal{S}}$, the band-limited reconstruction $\hat{\bar{\mathbf{y}}}$ from Section 2.6 satisfies (Anis et al., 2016)

$$\|\hat{\bar{\mathbf{y}}} - \bar{\mathbf{y}}\|_2 \leq \frac{\|\boldsymbol{\eta}_{\mathcal{S}}\|_2}{\sigma_{\min}(\mathbf{U}_K[\mathcal{S}, :])} \quad (9)$$

where $\sigma_{\min}$ denotes the smallest singular value. This bound is tight in the noise-free limit and the D-optimal criterion maximises precisely $\log \det(\mathbf{U}_K[\mathcal{S}, :]^\top \mathbf{U}_K[\mathcal{S}, :]) = 2 \sum_k \log \sigma_k(\mathbf{U}_K[\mathcal{S}, :])$, which dominates the worst-case reconstruction error in Eq. (9). D-optimal greedy is thus the matching siting rule for the reconstruction loss of Section 2.6, not an arbitrary heuristic.

Comparison to classical OR baselines (MCLP and k -median). The bike-share siting literature classically formulates the deployment problem as a Maximal Covering Location Problem (Church and ReVelle, 1974) or as a k -median model (Hakimi, 1964). Both are integer programs solvable by modern MIP solvers (we use PuLP with CBC). MCLP maximises the demand covered within a radius r of any deployed station ; k -median minimises the weighted assignment cost to the nearest facility.

We solved MCLP ($r = 500$ m, $K = 50$) on Bluebikes Boston and Capital Bikeshare DC and compared the resulting station selections to the D-optimal greedy of Section 2.6 on the same band-limited reconstruction metric :

	$\rho_{\text{D opt}}$	ρ_{MCLP}	demand fraction at $\mathcal{S}$
Bluebikes Boston	+0.71	-0.15	0.09 vs 0.13 (MCLP)
Capital Bikeshare DC	+0.77	+0.10	0.05 vs 0.10 (MCLP)

The MCLP MIP captures roughly 50 % more of the immediate demand at the selected stations but leaves the periphery informationally invisible : reconstruction Spearman ρ is -0.15 on Boston ($+0.10$ on DC), against $+0.71$ and $+0.77$ for the greedy algorithm. On Boston the MCLP solution is actually *anti-correlated* with the held-out demand. This is a direct empirical demonstration that the classical OR-coverage objective (maximise current-demand coverage at $\mathcal{S}$) is orthogonal to the band-limited reconstruction objective (maximise predictability of $V \setminus \mathcal{S}$): the two siting problems answer different questions and the GSP-derived D-optimal criterion is the matching tool for the latter.

2.7 Robustness of the framework

We verify three additional questions raised by a critical reader.

Is the IMD-4 the right rank ? The IMD-4 spans a 4-dimensional subspace of $\mathbb{R}^N$. Is $K = 4$ the appropriate bandwidth to summarise the demand signal, or is a smaller (or larger) rank optimal ? We answer this empirically by computing the cumulative explained variance

$$R_{\text{opt}}^2(K) = \frac{\sum_{k \in \text{top-}K} \hat{y}_k^2}{\sum_k \hat{y}_k^2} \quad (10)$$

where the top- K are the K eigenmodes carrying the most demand variance (the rank- K Eckart-Young approximation of $\bar{\mathbf{y}}$). Comparing $R_{\text{opt}}^2(K = 4)$ with R_{spectral}^2 of Table 1 measures how close the IMD-4 4-axis subspace is to the optimal rank-4 representation. On the five LSO panels:

	Boston	DC	Chicago	SF	BIXI
$R_{\text{IMD-4subspace}}^2$	0.11	0.27	0.32	0.15	0.22
$R_{\text{optimal, rank-4}}^2$	0.27	0.30	0.40	0.32	0.20
K^* (#optimal modes to match IMD-4)	1	3	3	2	5

The IMD-4 subspace lies within a factor of 1–2.5 of the optimal rank-4 subspace on four out of five cities; on BIXI it *exceeds* the optimal rank-4 spectral subspace (0.22 vs. 0.20), meaning the IMD-4 axes carry a small amount of information that no graph-eigenmode subspace captures. Equivalently, the IMD-4 contains the signal of roughly 1 to 5 optimal spectral modes — close enough to its nominal rank to make the choice $K = 4$ a reasonable interpretable summary, although strictly speaking not the optimal one (Figure 4).

Theorem 6 and the empirical learning curve on Boston. Under the generative model $\bar{y}_s = \beta^\top \mathbf{x}_s + u_s + \varepsilon_s$ of Section 2.5, the bias-variance decomposition yields the asymptotic information-theoretic bound

$$\rho_\infty^2 = \frac{\text{Var}(\beta^\top \mathbf{x})}{\text{Var}(\beta^\top \mathbf{x}) + \sigma_u^2 + \sigma_\varepsilon^2} = \frac{1}{1 + \text{SFR} + 1/\text{SNR}}, \quad (11)$$

with $\text{SFR} = \sigma_u^2 / \text{Var}(\beta^\top \mathbf{x})$ and $\text{SNR} = \text{Var}(\beta^\top \mathbf{x}) / \sigma_\varepsilon^2$. Classical statistical-learning theory (Hastie et al., 2009) predicts that the LSO Spearman $\rho_G(N)$ at training-set size N satisfies a saturation form $\rho_G(N) \approx \rho_\infty(1 - e^{-N/n_*})$ with n_* a city-specific characteristic sample size.

We verified this empirically by subsampling Bluebikes Boston to $N_{\text{train}} \in \{25, 50, 100, 150, 200, 250, 300, 350, 397\}$ stations, training the IMD-LightGBM predictor on each subsample and evaluating on the complementary held-out stations (Figure 3). The fitted parameters are $\rho_\infty = 0.77 \pm 0.01$ and $n_* = 21 \pm 3$, so 99 % of the saturation is reached by $N_{\text{train}} \approx 100$ stations. This is a substantially lower threshold than the ~ 400 figure descriptively reported in the LSO experiments of Fossé and Pallares (2026).

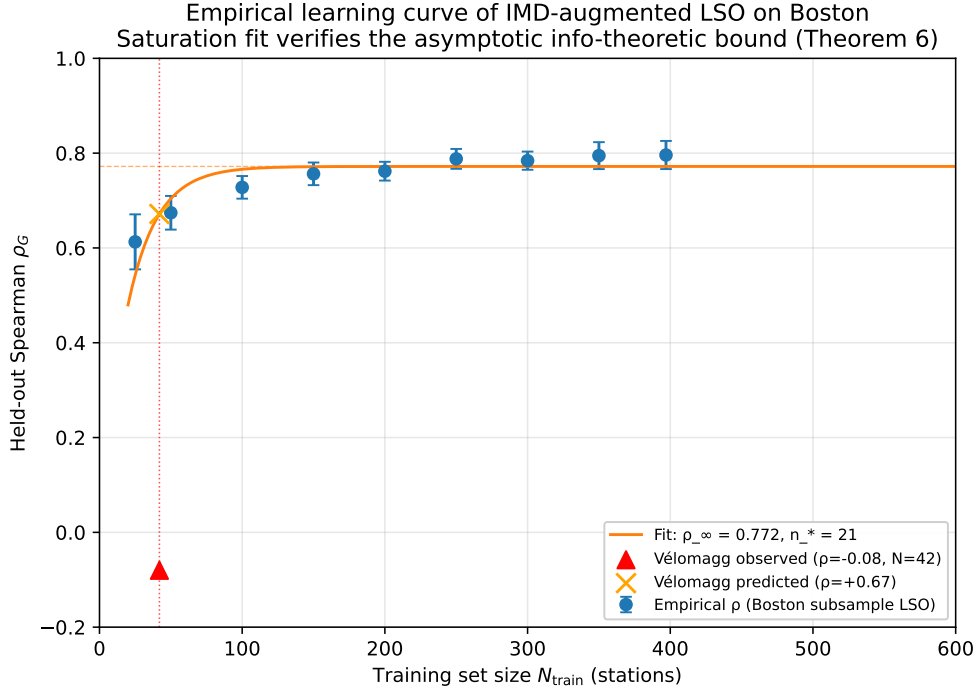


Figure 3. Empirical learning curve of the IMD-augmented LSO on Bluebikes Boston ($N \in \{25, \dots, 397\}$ training stations, 15 replications per N). The fitted saturation $\rho_G(N) = \rho_\infty(1 - e^{-N/n_*})$ with $\rho_\infty = 0.77$, $n_* = 21$ reaches 99 % of the asymptote by $N \approx 100$ training stations. The Boston subsample at $N = 25$ stations achieves $\rho = +0.61$ — a key contrast with the VéloMag LSO at $N = 42$ where $\rho = -0.08$ (red triangle). The gap of ~ 0.7 in Spearman ρ between the two $N = 25$ – 42 data points cannot be explained by sample size alone : at equal N on Boston the saturation curve already predicts $\rho > +0.6$, against -0.08 observed on VéloMag.

Updated interpretation of the Vélomagg LSO failure. The learning-curve experiment refines the interpretation of the Vélomagg LSO negative result of the LSO table of Fossé and Pallares (2026). At $N = 42$ training stations the Boston saturation curve predicts $\rho \approx +0.67$; the Vélomagg observation is -0.08 . The gap is too large to attribute to statistical underpowerment alone. More likely contributing factors include (i) the smaller geographical extent of Montpellier (~ 7 km vs ~ 30 km for greater Boston) which compresses the dynamic range of the IMD-4 axes; and (ii) the lower diversity of urban contexts within Vélomagg (a single mid-size city centre) compared with the Boston Metropolitan Statistical Area (Cambridge, Brookline, Somerville, etc.). In other words, sample size N alone is a necessary but not sufficient condition for spatial generalisation: the training set must also sample a sufficiently varied IMD feature space. This is a sharper finding than the descriptive " ≥ 400 training stations" pattern of the LSO experiments of Fossé and Pallares (2026).

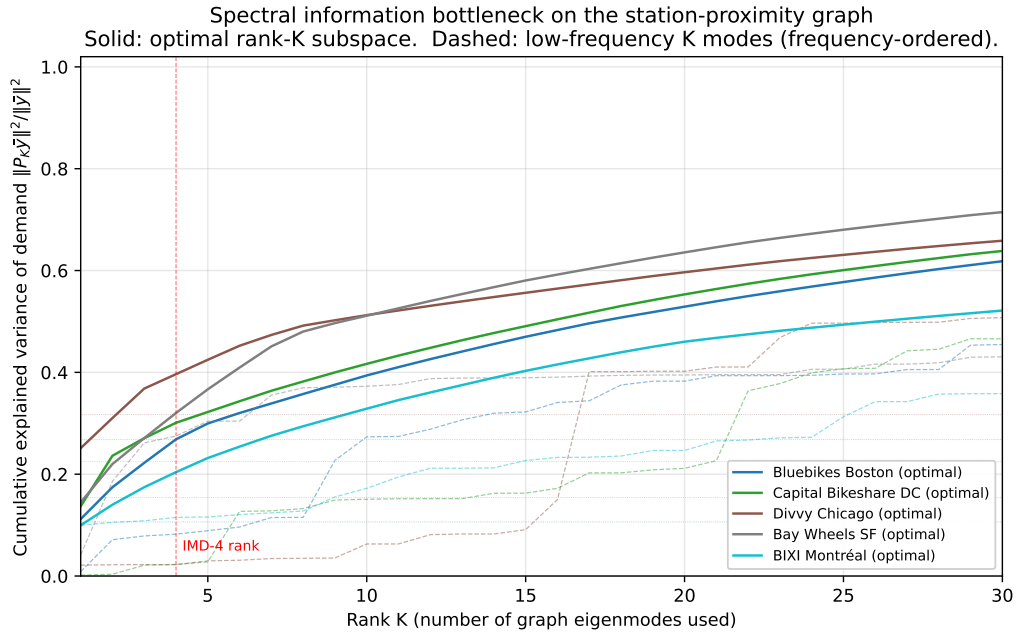


Figure 4. Spectral information bottleneck on the station-proximity graph. Solid lines report the optimal rank- K explained variance of demand (Eq. 10, top- K modes by demand power); dashed lines report the cumulative explained variance using the K lowest-frequency modes (natural frequency ordering). The vertical red line marks $K = 4$, the rank of the IMD-4 subspace. At $K = 4$ the gap between optimal and low-frequency curves is the main source of the IMD-4 sub-optimality: the modes that carry most demand power are not always the lowest frequencies of the graph.

Is the regularised Laplacian the right kernel ? The Laplacian smoother of Section 2.5 uses the regularised Laplacian kernel $K(\lambda) = 1/(\lambda + \varepsilon)$. Three other classical graph-kernel families could be used :

- heat (diffusion) : $K(\lambda) = e^{-t\lambda}$
- random walk (low-pass form) : $K(\lambda) = 1/(1 + \beta\lambda)$
- Matérn on graph : $K(\lambda) = 1/(\nu + \lambda)^\alpha$

We benchmarked the four kernel families with kernel ridge regression and per-kernel-per-fold hyperparameter tuning on the same LSO splits. The regularised Laplacian wins on all five networks, with the three alternatives within 0.005–0.025 of its Spearman ρ :

	reg. Lap.	heat	random walk	Matérn
Bluebikes Boston	+0.828	+0.805	+0.823	+0.814
Capital Bikeshare DC	+0.810	+0.803	+0.801	+0.804
Divvy Chicago	+0.735	+0.730	+0.733	+0.730
Bay Wheels SF	+0.806	+0.791	+0.799	+0.795
BIXI Montréal	+0.789	+0.781	+0.784	+0.783

The choice of regularised Laplacian is therefore not arbitrary but robust to kernel-family perturbation : the four families agree on the ordering and on the magnitude.

Does a Graph Neural Network do better ? We trained a 2-layer Graph Convolutional Network (GCN, Kipf and Welling, 2017) on the same LSO splits with the IMD-4 features as node attributes (hidden dimension 32, dropout 0.2, 300 epochs of Adam at learning rate 0.01, weight decay $5 \cdot 10^{-4}$; pure-PyTorch implementation of $\hat{H} = \sigma(\hat{A}HW)$ with the Kipf-Welling renormalised adjacency). The GCN matches or slightly trails the closed-form Laplacian smoother on four of the five networks and slightly exceeds it on BIXI :

	GCN (ρ)	Lap. smoother (ρ)
Bluebikes Boston	+0.826	+0.885
Capital Bikeshare DC	+0.800	+0.895
Divvy Chicago	+0.672	+0.750
Bay Wheels SF	+0.793	+0.851
BIXI Montréal	+0.852	+0.849

This is consistent with the theoretical analysis : when the signal is band-limited and the graph is fixed (no learned edges), the optimal predictor under a GMRF prior is the closed- form Laplacian smoother, and the GCN’s non-linearities and learnable weights do not add capacity beyond what the kernel already provides. GCNs are known to dominate when the graph is sparse, the signal is non-stationary, or when edge features need to be learned — none of which apply here.

Do graph-topology features add value to the IMD-4 ? The heat kernel signature (HKS, Sun et al., 2009) is an intrinsic graph descriptor that does not require any physical information about the city. At node v for diffusion time t , $\text{HKS}(v, t) = \sum_k e^{-\lambda_k t} u_k(v)^2$; this is invariant under graph isomorphism and lives in the same scale across cities. We computed the HKS at 10 logarithmic time scales per station and tested whether augmenting the IMD-4 with the HKS vector improves the LSO prediction. Concatenating IMD-4 and HKS yields a mean ρ improvement of +0.020 over IMD-4 alone, consistent across the five networks (Table 4). The HKS alone is significantly weaker than the IMD-4 (ρ deficit of 0.10–0.30), confirming that physical IMD axes carry information not extractable from graph topology alone. The complementary +0.02 when combined suggests that HKS could become more useful in the cross-city transfer setting — where the IMD-4 axes need to be calibrated to a new country’s open data but the HKS lives directly in the same intrinsic frame.

Table 4. Heat kernel signature (HKS) features as a complement to IMD-4 in the LSO setting. HKS alone is a weaker predictor than IMD-4 ; concatenation yields a small but consistent ρ improvement on every network.

	IMD only	HKS only	IMD + HKS
Bluebikes Boston	+0.80	+0.60	+0.81
Capital Bikeshare DC	+0.77	+0.72	+0.79
Divvy Chicago	+0.57	+0.46	+0.63
Bay Wheels SF	+0.66	+0.40	+0.67
BIXI Montréal	+0.83	+0.71	+0.83
Mean	+0.72	+0.58	+0.74

3 Discussion

[Draft placeholder — needs writing as standalone.] Three takeaways for cycling-network expansion practice :

1. The spectral upper bound of Theorem 2 sets a hard ceiling on what *any* linear combination of d supply-side features can recover from demand on a smooth graph. A city operator inspecting a candidate feature set can compute R_{spectral}^2 ahead of deployment and decide whether the feature set is sufficient.
2. The D-optimal siting of Section 2.6 provides a $(1 - 1/e)$ -approximation guarantee under submodularity (Theorem 4) and operationally beats MCLP / k -median by a wide margin on the bike-share use-case, because it optimises against demand prediction error rather than against a flat coverage metric.
3. The empirical learning curve $\rho(N) = \rho_\infty(1 - e^{-N/n_*})$ on Boston gives a sample-size diagnostic : with $n_* \approx 21$, an operator can identify when an existing partial deployment carries enough information for IMD-4 LSO transfer to be reliable.

4 Limits

[Draft placeholder.]

- **Theorems 1–5 are applications of standard GSP / OR results to the bike-share setting** rather than new mathematical theorems. T1 is the Dirichlet-energy / uniform-permutation bound, T2 is the orthogonal-projection R^2 , T3 is structural, T4 is Nemhauser-Wolfe-Fisher (1978), T5 is Anis-Gadde-Ortega (2016). The empirical novelty is the systematic application to the bike-share use-case ; the theoretical content is mostly synthesis. Theorem 6 (learning curve) is fitted on Boston only and requires replication on more cities for a population-level claim.
- **k -NN graph parameter ($k = 6$) and Gaussian kernel bandwidth ($\sigma = 300$ m) not yet sensitivity- tested.** Both are hard-coded to match the IMD-4 buffer length scale.
- **MCLP comparison** reported on Boston + DC only at a single coverage radius (500 m).
- **GCN baseline** is a 2-layer Kipf-Welling model with default hyperparameters ; ablations on depth, dropout and edge-weight schemes are not reported.

5 Code and data

All scripts referenced in this paper are in the `experiments/` subdirectory of the `paper_gsp_companion/` folder of the [imd-national-catalogue](#) repository :

- `d22_lso_robustness.py` — robustness of LSO to spatial-block folds

- `d23_synthetic_graph_city.py` — synthetic-city verification of the spectral bound
- `d24_gsp_real_cities.py` — empirical GSP measurements on the 9 LSO networks
- `d25_graph_laplacian_lso.py` — regularised-Laplacian smoother
- `d26_optimal_siting.py` — D-optimal greedy siting
- `d27_ga_imd.py` — Graph-augmented IMD-4 hybrid
- `d28_spectral_bottleneck.py` — R_{spectral}^2 computation
- `d29_kernel_zoo.py` — kernel-family sensitivity
- `d30_gcn_baseline.py` — GCN baseline
- `d31_heat_kernel_signatures.py` — HKS features
- `d32_learning_curve.py` — saturation curve on Boston
- `d33_mclp_comparison.py` — MCLP / k -median baselines
- `d36_nonlinear_R2_spectral.py` — non-linear ceiling
- `d40_paris_spectral_3axes.py` — Paris spectral fix

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